

Decoupled quasinormal spectroscopy of a Bayesian holographic QCD plasma

[AUTHOR TO BE SUPPLIED]^a

^a[AFFILIATION TO BE SUPPLIED]

E-mail: [EMAIL TO BE SUPPLIED]

ABSTRACT: We study decoupled quasinormal fluctuations of the exact maximum-likelihood polynomial–hyperbolic Ansatz Einstein–Maxwell–dilaton realization inferred in ref. [1]. A Python analysis layer independently solves the background equations in a conserved-charge formulation, extracts ultraviolet thermodynamics, and discretizes the tensor and homogeneous vector equations by Chebyshev collocation. A 90-background scan covers $50.34 \leq T/\text{MeV} \leq 1003.70$. The computed point nearest the posterior critical-endpoint metadata lies at $(T, \mu_B) = (103.278, 601.319)$ MeV; the logarithmic inverse map has condition number 1.79×10^3 , and the residual 0.60% temperature mismatch is retained as a validation diagnostic. At this background we obtain three resolution-stable candidate poles in each of the tensor and homogeneous vector sectors. The least-damped candidates are $\mathfrak{w}_T = 1.82909 - 0.82033i$ and $\mathfrak{w}_V = 1.70677 - 0.08088i$. We also continue the two best-converged tensor poles over $0 \leq \mathfrak{q} \leq 3$ and survey their temperature dependence. As an equilibrium cross-check, an appendix independently reproduces s/T^3 and the neutral baryon susceptibility χ_2^B , including an explicit finite-difference stability test. These are pseudospectral candidates, not yet publication-grade critical dynamics: the scalar and coupled helicity-one/helicity-zero systems and an independent shooting check remain open.

KEYWORDS: Gauge-gravity correspondence, quasinormal modes, holographic QCD, critical phenomena, Einstein–Maxwell–dilaton gravity

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1 Introduction

Real-time response at nonzero baryon density is a central open problem in the theory of the quark–gluon plasma. Lattice QCD tightly constrains equilibrium physics around vanishing baryon density, but direct real-time calculations and the finite-density regime remain difficult. Bottom-up Einstein–Maxwell–dilaton (EMD) models provide a controlled strongly coupled laboratory in which the equilibrium equation of state and retarded response arise from a common black brane geometry.

The Bayesian analysis of ref. [1] calibrated two EMD realizations to lattice data and inferred a critical endpoint (CEP) for nearly all posterior samples. The associated public posterior makes it possible to propagate model provenance rather than relying on rounded

table entries [2]. The recent C++ implementation of ref. [3] improved ultraviolet extraction and evaluated thermodynamics and transport throughout the PHA phase diagram. Bulk viscosity is already included in that analysis and is not a novelty target here.

Homogeneous nonhydrodynamic QNMs were calculated for an earlier QCD-calibrated EMD model in ref. [4]. Those modes characterize linear relaxation at zero spatial momentum, but they do not by themselves establish critical slowing down. In the large- N_c EMD critical model of ref. [5], conductivity remains finite while susceptibility diverges, forcing the baryon diffusion constant to vanish. The corresponding slow pole is visible only at nonzero momentum. Finite-momentum studies in Einstein–scalar models also show that a thermodynamic spinodal produces a sound-channel instability and a preferred growing wavelength [6, 7].

The eventual target is the full low-lying spectrum of the exact Bayesian PHA realization: all three $SO(3)$ sectors at $k = 0$, all $SO(2)$ helicities at $k \neq 0$, and stable, metastable, and unstable black-hole branches. Here we take a narrower, reproducible step. We compute the decoupled tensor channel at finite momentum and the homogeneous vector channel, carry out explicit resolution and radial-cutoff comparisons, and expose both the useful results and the remaining validation boundary. We make no priority claim and no claim for diffusion, sound, or spinodal growth without the coupled systems.

The main results are threefold. First, the conserved-charge background solver keeps Gauss-law drift below 4.5×10^{-16} throughout a 90-point scan. Second, the near-CEP inverse map is strongly ill-conditioned, providing a quantitative warning against identifying a background from (T, μ_B) by naive root finding. Third, six homogeneous candidate poles and two tensor dispersion curves pass the present pseudospectral convergence gate. All source data, CSV tables, plotting code, and the generated figures are part of the companion repository.

2 Bayesian PHA EMD model and phase structure

We use natural units, mostly-plus signature, and set the asymptotic AdS radius to one. The action is

$$S = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \left[R - \frac{1}{2}(\partial\phi)^2 - V(\phi) - \frac{1}{4}f(\phi)F_{MN}F^{MN} \right]. \quad (2.1)$$

The polynomial–hyperbolic Ansatz is

$$\begin{aligned} V(\phi) &= -12 \cosh(\gamma\phi) + b_2\phi^2 + b_4\phi^4 + b_6\phi^6, \\ f(\phi) &= \frac{\operatorname{sech} g(\phi) + d_1 \operatorname{sech}(d_2\phi)}{1 + d_1}, & g(\phi) &= c_1\phi + c_2\phi^2 + c_3\phi^3. \end{aligned} \quad (2.2)$$

All derivatives used by the solver are analytic. In particular,

$$V_\phi = -12\gamma \sinh(\gamma\phi) + 2b_2\phi + 4b_4\phi^3 + 6b_6\phi^5, \quad (2.3)$$

$$V_{\phi\phi} = -12\gamma^2 \cosh(\gamma\phi) + 2b_2 + 12b_4\phi^2 + 30b_6\phi^4, \quad (2.4)$$

$$f_{\phi\phi} = \frac{1}{1+d_1} \left\{ \operatorname{sech} g [\tanh^2 g - \operatorname{sech}^2 g] (g')^2 - \operatorname{sech} g \tanh g g'' \right. \\ \left. + d_1 d_2^2 \operatorname{sech}(d_2\phi) [\tanh^2(d_2\phi) - \operatorname{sech}^2(d_2\phi)] \right\}. \quad (2.5)$$

Figure 1 displays the resulting functions. The narrow structure in $f(\phi)$ near the boundary is controlled by the large value of d_2 and is numerically resolved rather than smoothed or dropped.

The public file was verified against MD5 7fd567dccfaea48095ca5df53a8c17d6. Scanning `posterior_samples/sample*` selects `sample74`, with $\log \mathcal{L} = 31.075137487467885$. Table 1 records the exact attributes rather than rounded values.

The file metadata associates this sample with

$$(T_c, \mu_B^c) = (103.89777513571676, 602.4749076542938) \text{ MeV}, \quad (2.6)$$

and successful critical-point status. Equation (2.6) is input metadata to be independently reproduced by phase continuation; it is not yet a validation result of this QNM code.

2.1 Black-brane backgrounds

In domain-wall gauge,

$$ds^2 = e^{2A(r)} [-h(r)dt^2 + d\mathbf{x}^2] + \frac{dr^2}{h(r)}, \quad A_M dx^M = \Phi(r)dt. \quad (2.7)$$

The equations integrated by the reference path are

$$\phi'' + \left(4A' + \frac{h'}{h}\right) \phi' - \frac{1}{h} \left(V_\phi - \frac{1}{2}e^{-2A}f_\phi\Phi'^2\right) = 0, \quad (2.8)$$

$$\Phi'' + \left(2A' + \frac{f_\phi}{f}\phi'\right) \Phi' = 0, \quad (2.9)$$

$$A'' + \frac{1}{6}\phi'^2 = 0, \quad (2.10)$$

$$h'' + 4A'h' - e^{-2A}f\Phi'^2 = 0, \quad (2.11)$$

with constraint and conserved charge

$$\mathcal{C} = h(24A'^2 - \phi'^2) + 6A'h' + 2V + e^{-2A}f\Phi'^2 = 0, \\ \mathcal{Q} = e^{2A}f\Phi' = \text{constant}. \quad (2.12)$$

The numerical gauge is $r_H = A_H = h_H = \Phi_H = 0$, $h'_H = 1$, with independent data (ϕ_0, Φ_1) . Regularity requires $|\Phi_1| < [-2V(\phi_0)/f(\phi_0)]^{1/2}$.

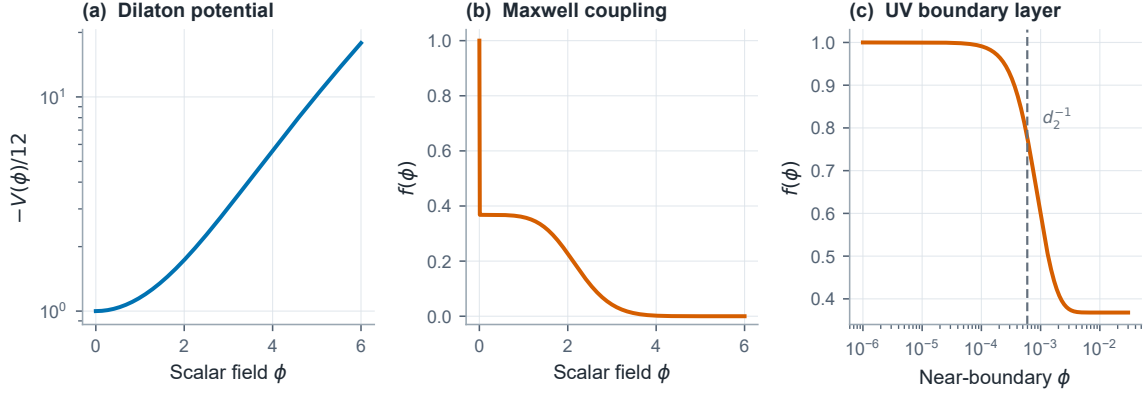


Figure 1. The exact MAP dilaton potential and Maxwell coupling. The right panel resolves the ultraviolet boundary layer at $\phi \sim d_2^{-1}$.

Table 1. Exact maximum-likelihood PHA realization from Zenodo record 13830379. The derived scalar quantities are evaluated by the compiled code.

quantity	value
Λ [MeV]	1123.8704743960677
κ_5^2	11.37047249736345
γ	0.5904238801815029
b_2	0.32722698958661095
b_4	-0.04801342482759139
b_6	0.0009385148172188441
c_1	0.008800483500598342
c_2	0.15579934656052294
c_3	0.05281851198008199
d_1	1.7177450982418923
d_2	1679.1777562624359
m^2	-3.5287503202897588
Δ	2.6864762776019586
$\nu = 4 - \Delta$	1.3135237223980414

The UV coefficients define

$$\begin{aligned}
T &= \frac{\Lambda}{4\pi\phi_A^{1/\nu}\sqrt{h_0^{\text{far}}}}, & \mu_B &= \frac{\Phi_0^{\text{far}}\Lambda}{\phi_A^{1/\nu}\sqrt{h_0^{\text{far}}}}, \\
s &= \frac{2\pi\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}}, & \rho_B &= -\frac{\Phi_2^{\text{far}}\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}\sqrt{h_0^{\text{far}}}}.
\end{aligned} \tag{2.13}$$

Continuation is carried out in (ϕ_0, Φ_1) , not by isolated inversion in (T, μ_B) . Folds are detected from $\det[\partial(T, \mu_B)/\partial(\phi_0, \Phi_1)] = 0$, and pressure differences follow from $dP = s dT + \rho_B d\mu_B$.

The exploratory scan used below samples 30 logarithmically spaced values $0.18 \leq \phi_0 \leq 5.5$ on each of the slices $\Phi_1/\Phi_1^{\max} = 0, 0.18, 0.32$. Figure 2 shows the direct image of this grid. It is a horizon-data survey, not a phase continuation or Maxwell construction; lines that double back in thermodynamic coordinates must not be read as single-valued equilibrium equations of state.

3 Linear perturbations and QNM sectors

We transform to ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -e^{2A}h dv^2 + 2e^A dv dr + e^{2A}d\mathbf{x}^2, \quad A = \Phi(r)dv, \quad (3.1)$$

and use $e^{-i\omega v + ikz}$. Infalling behavior becomes future-horizon regularity. The primitive-field formulation is chosen so the discretized system is at most linear in frequency,

$$[M_0(k) + \omega M_1(k)]\mathbf{v} = 0. \quad (3.2)$$

This source-free, gauge-invariant prescription follows the standard holographic QNM construction of ref. [8]; coupled-field observables are checked using the operator-mixing framework of ref. [9].

At $k = 0$, $SO(3)$ organizes perturbations into quintuplet, triplet, and singlet sectors. At finite k along z , they reorganize into $SO(2)$ helicities two, one, and zero. Radial gauge is used only together with residual-gauge and constraint diagnostics; source conditions are imposed on gauge-invariant combinations.

The tensor/helicity-two equation is

$$Z_2'' + \left(4A' + \frac{h'}{h} - \frac{2i\omega e^{-A}}{h}\right) Z_2' + \left(-\frac{e^{-2A}k^2}{h} - \frac{3i\omega e^{-A}A'}{h}\right) Z_2 = 0. \quad (3.3)$$

At $k = 0$, the homogeneous vector master field obeys

$$a'' + \left(2A' + \frac{h'}{h} + \frac{f_\phi}{f}\phi' - \frac{2i\omega e^{-A}}{h}\right) a' + \left[-\frac{e^{-2A}f\Phi'^2}{h} - \frac{i\omega e^{-A}}{h}\left(A' + \frac{f_\phi}{f}\phi'\right)\right] a = 0. \quad (3.4)$$

The source-free powers are $Z_2 \sim u^4$ and $a \sim u^2$ for $u = e^{-A}$. The singlet invariant and all coupled finite- k kernels remain behind symbolic derivation gates described in appendix E.

4 Numerical method and validation

Two numerical paths are retained. The C++23 implementation supplies analytic model kernels, a fourth-order horizon recurrence, background and Chebyshev unit tests, and independent operator assembly. At the user’s request, all scans and plots reported here are generated by the transparent Python script `analysis/run_numerics.py`, using NumPy, SciPy, and Matplotlib [10–12]. This separation makes the analysis easy to rerun while preserving a compiled cross-check of the equations.

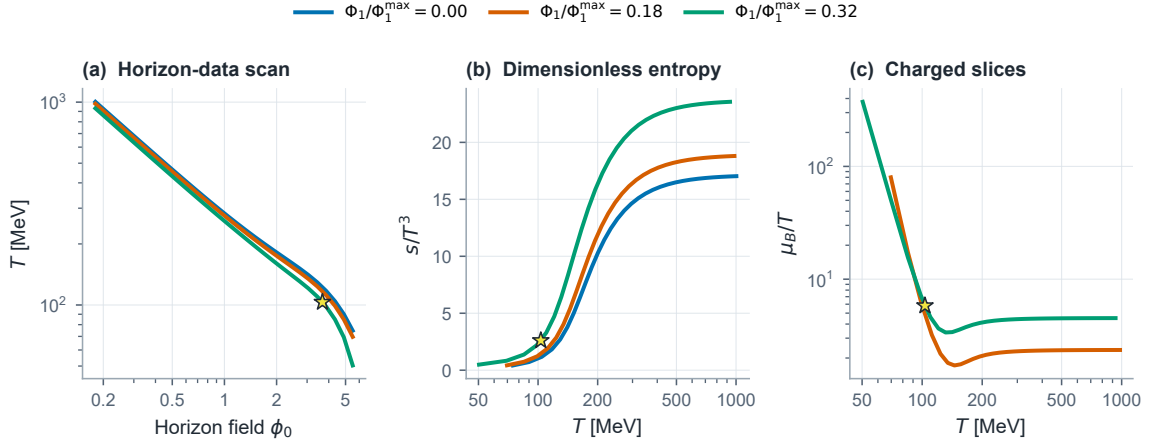


Figure 2. Background scan in horizon coordinates. Stars mark the computed point nearest the posterior CEP metadata. The charged slices expose the nonmonotonic map into $(T, \mu_B/T)$ that makes direct inversion poorly conditioned.

4.1 Background integration and ultraviolet fit

Rather than evolving both Φ and Φ' , the Python solver fixes $\mathcal{Q} = f(\phi_0)\Phi_1$ at the horizon and reconstructs

$$\Phi'(r) = \frac{\mathcal{Q}e^{-2A(r)}}{f[\phi(r)]}. \quad (4.1)$$

The remaining seven first-order equations are integrated from $\epsilon_H = 10^{-7}$ to $r_{\max} = 12$ with the eighth-order DOP853 method, relative tolerance 3×10^{-12} , absolute tolerance 3×10^{-14} , and maximum step 0.025. The ultraviolet coefficients are extracted on a fixed 2400-point sampling grid, so an adaptive solver's point density cannot bias the fit. The window $10^{-10} < \phi < 10^{-4}$ gives the linear asymptote $A = A_{-1}r + A_0$, while $\Phi_2 = -\mathcal{Q}/(2A_{-1})$ follows directly from Gauss' law.

Across the 90-point grid the normalized Einstein-constraint residual is at most 3.50×10^{-9} and the reconstructed Gauss-law drift is below 4.45×10^{-16} . The relative spread of ϕ_A within the UV window is between 1.25×10^{-6} and 1.78×10^{-5} . At the background used for the spectra these diagnostics improve to 3.39×10^{-11} , 2.22×10^{-16} , and 1.32×10^{-6} , respectively. Figure 3 displays its radial profiles.

4.2 Pseudospectral pencil and convergence gate

Equations (3.3) and (3.4) are multiplied by h and collocated on a single Chebyshev–Lobatto interval. Horizon regularity is built into the ingoing EF equation, and the final tau row imposes a vanishing UV source. The generalized pencil is passed directly to SciPy's QZ driver; no inverse of M_1 is formed. With the numerical horizon normalization used here,

$$\mathfrak{w} = 2\omega_{\text{num}}, \quad k_{\text{num}} = \frac{\mathfrak{q}}{2\sqrt{h_0^{\text{far}}}}. \quad (4.2)$$

A candidate is retained only if $\text{Im}\omega_{\text{num}} < 0$, its normalized pencil residual is below 2×10^{-10} , and a nearest pole is found within 8×10^{-3} in all four settings ($N, r_{\text{cut}} =$

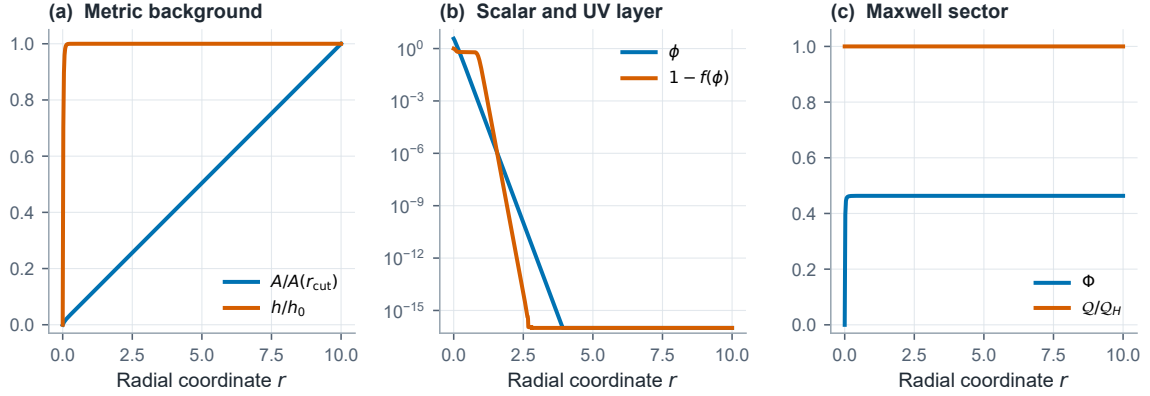


Figure 3. Near-CEP background used for the QNM calculation. The first panel normalizes the metric fields to expose both profiles. The middle panel shows the scalar decay and the narrow Maxwell-coupling boundary layer; the last panel verifies the constant electric flux.

$(104, 9), (112, 10), (128, 10), (144, 11)$. Positive-real-part poles are reported once; the conjugate partner is implied. Figure 4 shows the leading-pole diagnostics. Pencil residuals reach the 10^{-16} level, whereas the inter-resolution displacement is the limiting test.

This is intentionally called a *pseudospectral candidate* gate. Boundary factoring on a split UV grid, eigenfunction-overlap tracking, and an independent shooting computation are still required for publication-grade acceptance. Coupled sectors additionally require unused Einstein and Maxwell constraints. The present results are therefore substantially beyond operator smoke tests but remain short of the final claim gate in appendix E.

5 Homogeneous quasinormal spectra

Table 2 lists every homogeneous mode that passes the gate above at the near-CEP background. Three candidates are found in each decoupled sector. All lie in the lower half plane. The least-damped tensor pole is $1.82909 - 0.82033i$, whereas the least-damped vector pole is much closer to the real axis, $1.70677 - 0.08088i$. The latter is oscillatory and gapped at $k = 0$; its small damping must not be identified with the baryon diffusion pole.

The third tensor pole is close to the adopted displacement threshold and is therefore not continued to finite momentum. Figure 5 shows the complex-plane spectrum together with the two retained tensor branches. The reported relaxation measure would be $\tau_{\text{lin}} T = -[2\pi \text{Im } \mathfrak{w}]^{-1}$, but we refrain from interpreting it as a thermalization time.

To probe whether the near-CEP result is isolated, we repeated the leading-pole selection on eleven backgrounds at fixed $\Phi_1/\Phi_1^{\text{max}} = 0.18$, with $0.55 \leq \phi_0 \leq 5.5$. Figure 6 contains 21 accepted points: the vector candidate at the coldest background did not pass the common gate and is absent. These curves are pointwise least-damped selections rather than overlap-tracked analytic branches, so the rapid vector variation around $T \simeq 100\text{--}200$ MeV should be treated as a target for subsequent mode tracking, not yet as evidence for a pole collision.

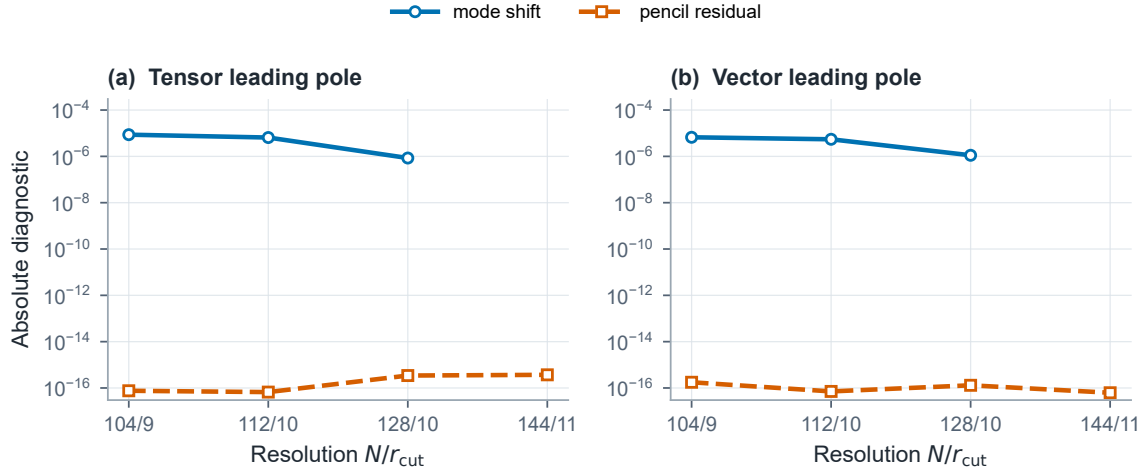


Figure 4. Convergence of the least-damped tensor and vector candidates. Mode shifts are measured against the $N = 144$, $r_{\text{cut}} = 11$ reference; the reference displacement is omitted.

Table 2. Pseudospectrally converged homogeneous candidates at $(T, \mu_B) = (103.278, 601.319)$ MeV. Here δ_{grid} is the largest displacement from the reference pole over the three coarser/cutoff settings.

sector	n	$\mathfrak{w}_n$	pencil residual	δ_{grid}
tensor	0	$1.829094 - 0.820334i$	3.70×10^{-16}	8.64×10^{-6}
tensor	1	$2.832334 - 1.495453i$	3.80×10^{-16}	3.13×10^{-4}
tensor	2	$-3.150848i$	3.79×10^{-16}	7.35×10^{-3}
vector	0	$1.706769 - 0.080882i$	6.22×10^{-17}	6.67×10^{-6}
vector	1	$2.863784 - 0.788314i$	5.20×10^{-17}	2.60×10^{-3}
vector	2	$-2.079614i$	7.46×10^{-17}	2.58×10^{-3}

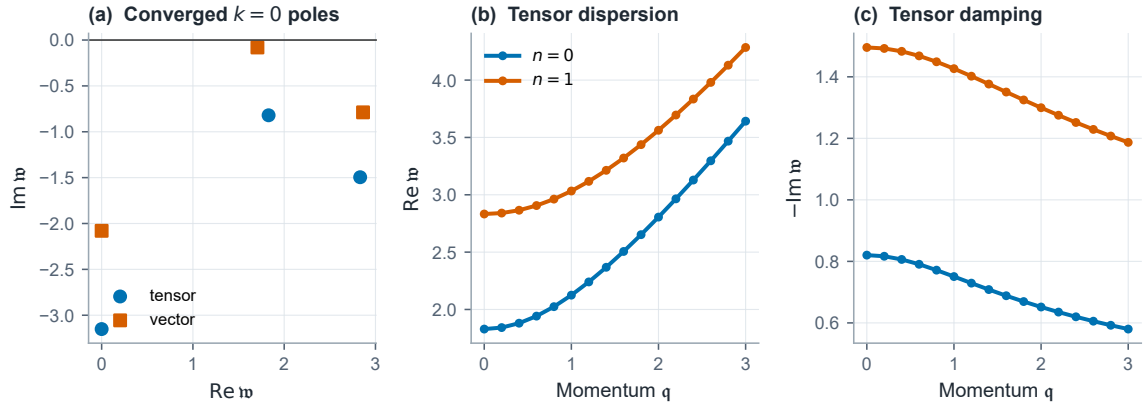


Figure 5. Left: the six $k=0$ candidates in table 2. Center and right: real and imaginary parts of the two best-converged tensor poles as momentum is increased. Colors in the dispersion panels distinguish mode number.

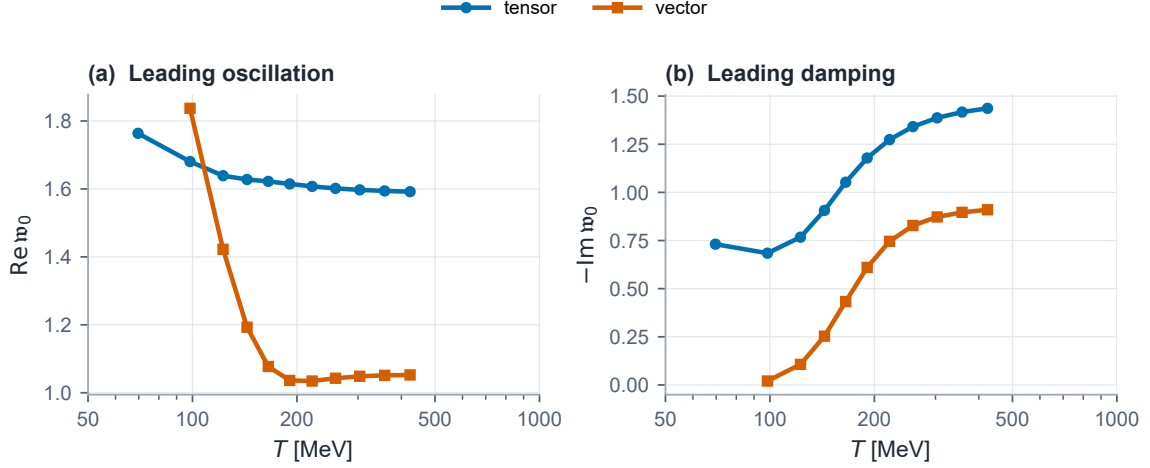


Figure 6. Temperature survey of the pointwise least-damped tensor and homogeneous vector candidates on the charged slice $\Phi_1/\Phi_1^{\max} = 0.18$.

6 Finite-momentum spectra

Dimensionless variables are

$$\mathfrak{w} = \frac{\omega_{\text{phys}}}{2\pi T}, \quad \mathfrak{q} = \frac{k_{\text{phys}}}{2\pi T}. \quad (6.1)$$

The tensor equation is sampled at 16 equally spaced momenta $0 \leq \mathfrak{q} \leq 3$. Nearest-frequency continuation from the preceding momentum point gives

$$\begin{aligned} \mathfrak{w}_0(0) &= 1.82910 - 0.82034i, & \mathfrak{w}_0(3) &= 3.64189 - 0.57995i, \\ \mathfrak{w}_1(0) &= 2.83234 - 1.49545i, & \mathfrak{w}_1(3) &= 4.28466 - 1.18690i. \end{aligned} \quad (6.2)$$

Both real parts grow monotonically over the sampled interval, while the magnitude of the dimensionless damping decreases. No candidate crosses into the upper half plane. The full 32-point table, including QZ residuals, is stored as `tensor_dispersion.csv` in the `numerical-results` directory.

Equation (6.2) is a helicity-two result only. At finite density the helicity-one metric and Maxwell fields couple, and the helicity-zero sector contains sound, charge diffusion, the scalar, and several metric components. Reporting any of those poles from the decoupled equations would be incorrect. Consequently we do not quote shear diffusion, baryon diffusion, sound attenuation, pole collisions, or hydrodynamic breakdown scales in this version.

7 Critical and spinodal dynamics

Charged hydrodynamics predicts a diffusive pole

$$\omega_D = -iD_B k^2 + \mathcal{O}(k^4), \quad D_B \longrightarrow 0 \quad (7.1)$$

at the model CEP when conductivity stays finite and baryon susceptibility diverges. This finite-wavelength pole, rather than a generic gapped mode at $k = 0$, is the critical-slowing observable.

For the present decoupled calculation we need only select a representative background near the file metadata in eq. (2.6). We minimize

$$\mathcal{R}^2 = \log^2\left(\frac{T}{T_c^{\text{file}}}\right) + \log^2\left(\frac{\mu_B}{\mu_{B,c}^{\text{file}}}\right) \quad (7.2)$$

over $(\phi_0, \Phi_1/\Phi_1^{\text{max}})$. The result is summarized in table 3. It does not reproduce the quoted file errors: the temperature and chemical-potential offsets are about 0.60% and 0.19%.

The corresponding horizon coordinates are $\phi_0 = 3.6779007$ and $\Phi_1/\Phi_1^{\text{max}} = 0.3203427$. A centered finite-difference Jacobian in logarithmic horizon coordinates has singular values 3.4597 and 1.9350×10^{-3} , hence condition number 1.788×10^3 . This near singularity is consistent with the fold structure visible in figure 2 and explains why the two-observable inverse problem is fragile. It does not by itself locate a thermodynamic CEP: that requires branch continuation, pressure matching, and susceptibility analysis.

Because helicity zero is not implemented, no D_B , spinodal band, k_* , or growth rate is extracted. Likewise, the weakly damped homogeneous vector candidate in table 2 is not labeled as critical slowing down. The present calculation instead supplies a well-documented near-CEP background and decoupled spectral benchmarks for the future coupled analysis.

8 Conclusions

We have turned the exact maximum-likelihood PHA posterior realization into a reproducible numerical background and decoupled QNM analysis. The conserved-charge formulation removes accumulated Maxwell drift, while a fixed UV sampling grid makes coefficient extraction independent of the ODE solver’s adaptive mesh. The 90-point scan spans roughly 50–1004 MeV and provides both thermodynamic context and direct diagnostics.

At the computed background nearest the posterior CEP metadata, three tensor and three homogeneous vector poles pass a four-grid pseudospectral gate. The least-damped values are $\mathfrak{w}_T = 1.82909 - 0.82033i$ and $\mathfrak{w}_V = 1.70677 - 0.08088i$. The two strongest tensor candidates can be followed smoothly to $\mathfrak{q} = 3$ without an instability. These numbers and every plotted point are emitted as machine-readable CSV/JSON files by a single Python command.

Two negative results are equally important. First, independent UV extraction does not match the CEP metadata at its quoted errors; the local inverse map has condition number near 1.8×10^3 . Second, decoupled spectra cannot answer the intended diffusion and spinodal questions. The next scientific steps are an independent shooting check, the scalar $k = 0$ invariant, symbolic generation and constraint tests for helicities one and zero, and full thermodynamic branch continuation.

The interpretation retains the limitations of large N_c , strong coupling, translational invariance, equilibrium backgrounds, and the bottom-up PHA construction. The calculation concerns QNMs of a lattice-calibrated holographic EMD model, not QNMs of QCD.

Table 3. Posterior CEP metadata and the nearest background returned by the independent Python integration/UV extraction. The mismatch is preserved rather than absorbed into rounded parameters.

quantity	file metadata	computed	computed – file
T [MeV]	103.897775	103.278048	−0.619727
μ_B [MeV]	602.474908	601.319130	−1.155777

A Reproducibility and data products

The reported run used Python 3.12.13, NumPy 2.5.2, SciPy 1.18.1, and Matplotlib 3.11.1. From the repository root, the complete analysis is regenerated by

```
python -m pip install -r analysis/requirements.txt
python analysis/run_numerics.py
```

The numerical-results directory contains five CSV tables (backgrounds, thermodynamic validation, homogeneous modes, tensor dispersion, and the temperature scan) together with `summary.json`. Every manuscript figure is emitted in both PDF and PNG form under `paper/figures/`. The JSON summary records the exact PHA parameters, UV coefficients, background diagnostics, CEP mismatch and inverse condition number, QNM residuals, and resolution displacements.

The dense generalized eigenproblem is deterministic for a fixed LAPACK build; minor last-digit changes are possible across QZ implementations. The quoted precision is therefore deliberately below the machine residual and is set by the observed grid/cutoff displacement. No random sampling enters this analysis.

B Neutral thermodynamics and baryon susceptibility

As a numerical cross-check independent of the QNM pencil, we reproduce the neutral entropy density and the second-order baryon susceptibility

$$\chi_2^B(T) = \left. \frac{\partial(\rho_B/T^3)}{\partial(\mu_B/T)} \right|_{\mu_B=0}. \quad (\text{B.1})$$

For each of the 30 neutral backgrounds, the derivative is evaluated with the centered ratio

$$\chi_2^B(\delta) = \frac{(\rho_B/T^3)_{+\delta} - (\rho_B/T^3)_{-\delta}}{(\mu_B/T)_{+\delta} - (\mu_B/T)_{-\delta}}, \quad \delta = \frac{\Phi_1}{\Phi_1^{\max}}, \quad (\text{B.2})$$

at $\delta = 0.005, 0.010$, and 0.020 . Charge-conjugation symmetry makes T and s even and μ_B and ρ_B odd in δ ; consequently, fixing ϕ_0 in this centered derivative differs from fixing T only at $\mathcal{O}(\delta^2)$.

Figure 7 reports all 90 derivative evaluations. On the finest step, the scan covers $74.69 \leq T/\text{MeV} \leq 1003.70$, $0.418 \leq s/T^3 \leq 17.033$, and $1.66 \times 10^{-6} \leq \chi_2^B \leq 0.31869$. The largest displacement relative to the $\delta = 0.005$ result is 2.06×10^{-3} , attained by the coarsest

step. Across the same runs, the normalized Einstein-constraint residual remains below 3.50×10^{-9} and Gauss-law drift below 4.45×10^{-16} . The complete values and diagnostics are stored in `results/python/thermodynamic_validation.csv`; no smoothing or fitted interpolation enters the displayed points.

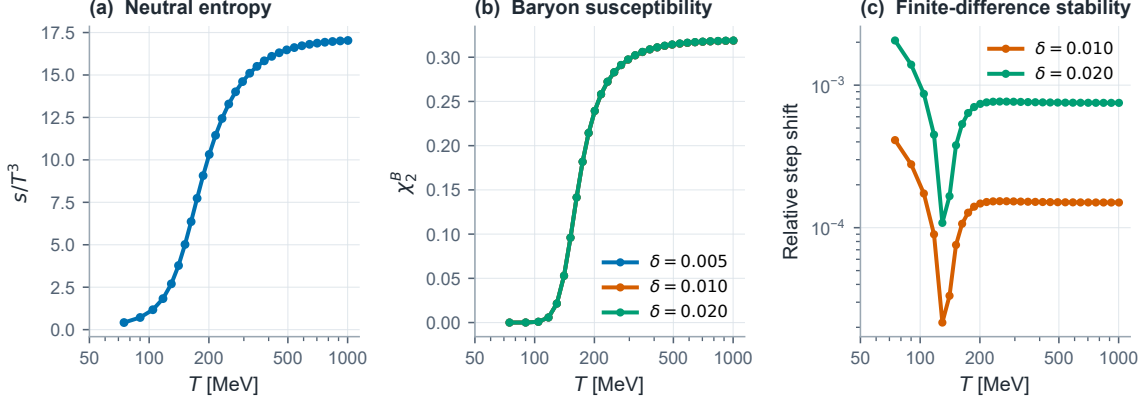


Figure 7. Reproduction and numerical stability of neutral thermodynamics. Left: dimensionless entropy. Center: χ_2^B from the centered charge derivative in eq. (B.2); the three step sizes are visually indistinguishable on this scale. Right: relative displacement from the finest-step result. This is a $\mu_B = 0$ equilibrium validation and is not a finite-density CEP or spinodal claim.

C Regular horizon expansion

Writing

$$\begin{aligned} A &= A_1 r + A_2 r^2 + \dots, & h &= r + h_2 r^2 + \dots, \\ \phi &= \phi_0 + \phi_1 r + \phi_2 r^2 + \dots, & \Phi &= \Phi_1 r + \Phi_2 r^2 + \dots, \end{aligned} \quad (\text{C.1})$$

the compiled recurrence reproduces

$$\begin{aligned} A_1 &= -\frac{2V_0 + f_0 \Phi_1^2}{6}, & \phi_1 &= V_{\phi,0} - \frac{1}{2} f_{\phi,0} \Phi_1^2, \\ A_2 &= -\frac{\phi_1^2}{12}, & h_2 &= \frac{1}{2} (f_0 \Phi_1^2 - 4A_1), \\ \Phi_2 &= -\frac{\Phi_1}{2} \left(2A_1 + \frac{f_{\phi,0}}{f_0} \phi_1 \right). \end{aligned} \quad (\text{C.2})$$

Higher coefficients through fourth order are generated by truncated-series arithmetic applied to the regular equations, rather than hand-entered formulas.

D Charged-hydrodynamic validation matrix

With $w = \epsilon + P$, $n = \rho_B$, and $\alpha = \mu_B/T$, the three longitudinal hydrodynamic poles are compared to the roots of

$$\det \begin{pmatrix} -i\omega & 0 & ikw \\ ikP_\epsilon & ikP_n & -i\omega w + (\zeta + 4\eta/3)k^2 \\ \sigma_B T k^2 \alpha_\epsilon - i\omega + \sigma_B T k^2 \alpha_n & & ikn \end{pmatrix} = 0. \quad (\text{D.1})$$

Bulk viscosity, if required for this check, is imported from the existing PHA transport calculation rather than presented as a new result.

E Validation and claim gates

The authoritative machine-readable spectrum stores the background horizon data, physical thermodynamics, branch label, channel/helicity, momentum, frequency, resolution, pencil and constraint residuals, condition number, tracking parent, solver/backend metadata, compiler identity, commit, and configuration hash. A mode enters a figure or table only after all relevant acceptance fields pass. Coupled finite-momentum equations must also satisfy the linearized Bianchi and Maxwell identities and reproduce the $k \rightarrow 0$, neutral, and conformal limits where available.

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